Ganga Nair B

Website | Email (Work) | Email (Personal) | +91 8138091164 | GitHub | Google Scholar | LinkedIn

Robotics researcher specializing in **learning-based control** for dynamic robotic systems, combining **optimal control**, **reinforcement learning**, and **physics-informed learning** to build interpretable controllers.
Currently pursuing an **M.Tech at Indian Institute of Science**, developing adaptive locomotion and structured dynamics models for **quadruped robots**, with focus on real-time control and sim-to-real transfer.

Robotics Research

Quadruped Locomotion using Central Pattern Generators

February 2026 – Current *Kyoto University*

Advisors: Prof. Jun Morimoto, Prof. Shishir N Y Kolathaya

- Developing a locomotion framework that optimizes CPG parameters given a morphology for a terrain, and payload.
- Using Conditional Bayesian Optimization to enable online parameter optimisation for energy-efficient gait adaptation.

Physics-Guided Generative Dynamics Modeling

August 2025 – March 2026 *Indian Institute of Science*

Advisors: Prof. Shishir N Y Kolathaya

- Built a dynamics learning framework that models conservative rigid-body mechanics using Lagrangian Neural networks to preserve energy-consistent inertial dynamics and models uncertain, non-conservative interaction effects using Conditional Flow Matching (CFM) to capture its multi-modal nature.
- Validated on Unitree Go1 and Unitree G1ub.io/STRIDE/[Website]
- `[Conference Paper]` Prakrut K., Ganga N. B., Shishir K., **STRIDE: Structured Lagrangian and Stochastic Residual Dynamics via Flow Matching**, *IEEE/RSJ IROS 2026* (Under review). [Paper]
- `[Workshop Paper]` Prakrut K., Ganga N. B., Shishir K., **STRIDE: Structured Lagrangian and Stochastic Residual Dynamics via Flow Matching**, *IEEE ICRA 2026: Workshop on the Path Towards Generalizable Contact-Rich Robotics: Control and Representation* (Accepted).

Physics-Informed ML for Safe and Optimal Control

October 2025 – February 2026 *Indian Institute of Science*

Advisors: Prof. Shishir N Y Kolathaya, Prof. Somil Bansal

- Worked on the co-optimization of safety and performance as a robust state-constrained optimal control problem.
- The resultant value function is approximated using a novel physics-informed machine learning framework. Additionally, a conformal prediction-based verification strategy is introduced to quantify the learning errors.
- `[Journal]` Manan T., Aditya S., Ganga N. B., Pushpak J., Shishir K., Somil S., **A Robust Physics-Informed Machine Learning Framework for Safe and Optimal Control of Autonomous Systems**, *International Journal of Robotics Research* (Under review).

Real-Time Gait Adaptation for Quadruped Locomotion

March 2025 – December 2025 *Indian Institute of Science*

Advisors: Prof. Shishir N Y Kolathaya

Education

M.Tech. in Robotics and Autonomous Systems

Indian Institute of Science, Bengaluru
July 2024 – June 2026

- **CGPA:** 8.80/10
- **Thesis:** Gait adaptation for efficient quadruped locomotion
- **Coursework:** Nonlinear Control, Reinforcement Learning, Robotic Perception, Motion Planning, Swarm Robotics, Machine Learning

B.Tech. in Mechanical Engineering

National Institute of Technology, Calicut
July 2018 – June 2022

- **CGPA:** 9.19/10
- **Thesis:** Dynamic Vibration Absorber with MR Damper [Link]
- **Coursework:** Control Systems, Nonlinear dynamics, Embedded Systems, Solid Mechanics, Machine Design, Manufacturing

CBSE Class XII

Bhavans Vidya Mandir, Eroor
June 2017 – May 2018

- **Score:** 96%

Robotics Projects

- **Safe Swarm Navigation using Control Barrier Functions**: Developed a CBF-based control layer for quadrotor swarm navigation in constrained environments. Implemented a priority-based CBF filter on a base formation controller which deforms the formation to allow navigation through constrained spaces.
- **Online iLQR with Neural Network Dynamics**: Integrated iLQR with an online-trained neural model to achieve

- Developed an MPC-based gait selection framework for terrain- and task-aware locomotion.
- Integrated a dreamer to provide warm-start policy and reward priors to an MPC framework. It was trained using supervised learning over a gait-conditioned Reinforcement Learning policy.
- Deployed continuous gait transitions on Unitree Go1 in simulation and hardware for improved energy use and velocity tracking.
- `[Poster]` Presented as a late-breaking result at the **2025 IEEE-RAS Humanoids Conference**. [Website]
- `[Conference Paper]` Ganga N. B., Prakrut K., Shishir K., **Real-Time Gait Adaptation for Quadrupeds Using Model Predictive Control and Reinforcement Learning**, *Indian Control Conference 2025*. [Website] [Paper]

Patent

- **Magneto-rheological Fluid-Based Dynamic Vibration Absorber**, Indian Patent Application No. 202241036483, filed June 24, 2022; (Patent pending).

Work Experience

Researcher
Learning Machines Group, Kyoto University *Feb 2026 – Current*

- Working on a joint project with IISc on morphology-independent quadruped locomotion framework using central pattern generators under the guidance of Prof. Jun Morimoto and Prof. Satoshi Yamamori.

Project Engineer
ExxonMobil Services & Technology, Bengaluru *July 2022 – August 2024*

- Led engineering and procurement for **10 refinery projects** with total scope up to **$25M**, coordinating civil, piping, and instrumentation stakeholders.
- Recognized within the company for introducing impactful innovations that improved project cost and schedule efficiency including deployment of 3D scanning in hazardous underground waste pits to eliminate the need for human entry

Execution Engineer Intern
ExxonMobil Services & Technology, Bengaluru *March 2021 – May 2021*

- Worked on developing the Project Plan and associated tools for Regulatory Project.

Skills

Python | C++ | MATLAB | PyTorch | JAX | NumPy | OpenCV
ROS | Gazebo | Isaac Gym | Isaac Sim | PyBullet | MuJoCo
Git | Linux | Simulink | SolidWorks | Ansys | Docker

model-based control under unknown nonlinear dynamics. Demonstrated performance on MountainCar and CartPole without offline dynamics pretraining. [Link]

- **Comparison and SLAM Implementation**: Implemented metric-semantic SLAM, trajectory evaluation, and object-level segmentation in indoor environments using an Intel RealSense depth camera. Performed a comparative study between Kimera and RTAB-MAP approaches. [Link]
- **Leader-Follower Formation Control**: Developed leader-follower formation control for TurtleBot 4 on ROS2 and Gazebo, including real-time control of the leader robot to ensure follower convergence to the desired trajectory. [Link]

Awards & Recognition

- Selected for the **India Japan Circulation of Talented Youths in Science Program (LOTUS)** 2025.
- Received the **Kanako Miura Award Grant** at the **2025 IEEE-RAS Humanoids Conference**.
- Awarded the **Reliance Postgraduate Scholarship** (top 100 postgraduate students in India, 2024).
- Received a **fellowship from Axis Bank** for the first year of the master's program.
- Earned **ExxonMobil recognitions** for execution workflow and safety identity initiatives.

Other Activities

- Provided academic support through volunteer work with **Make a Difference** and **Science for Rural India**.
- Delivered technical and non-technical talks and received awards for public speaking and presentations.
- Enjoy writing and blogging on technical and general-interest topics.